Sound: pitch and amplitude

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a drawing or explanation of a longitudinal wave to distinguish pitch from sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pitch” and “amplitude” to explain your answer. Sentence starters are optional.

pitch: the perceived highness or lowness of a sound

amplitude: maximum displacement from equilibrium for the wave

longitudinal: a disturbance parallel to the wave’s travel direction

01 Warm-up: bring back an earlier idea

A wave enters a medium where its speed is **3** times as large, at a small angle below the critical angle. Does the transmitted ray bend toward or away from the normal?

Word help: medium = the material through which a mechanical wave travels

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

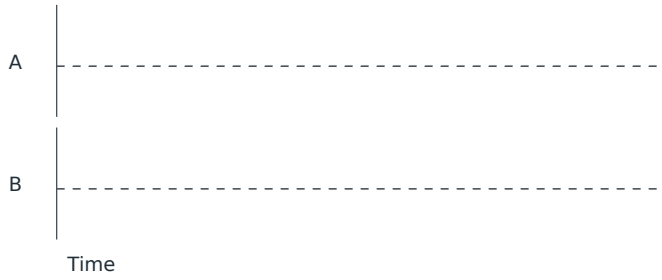
02 Worked example: explain the prediction

Reference: $T = 1/f$; $v = f\lambda$.
Higher f generally means higher pitch. Amplitude affects intensity with the other parts of the setup kept the same, not frequency.

Worked reasoning: A tuning fork makes **440 cycles** each second, so its frequency is **440 Hz**. A louder version can have the same frequency but greater amplitude.

Your turn: Draw two displacement-time traces with equal periods and different amplitudes.

Sentence starter: Both ... . They differ in ... because ... .

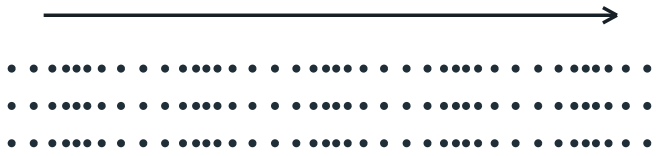


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?

Sentence starter: I notice ... . This supports ... because ... .



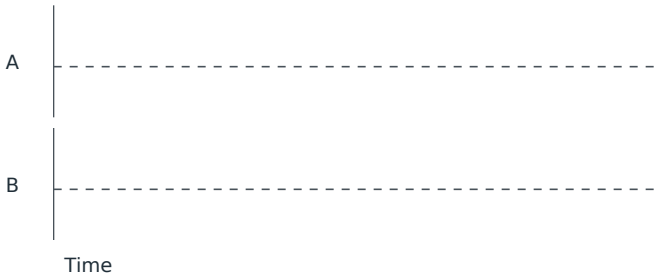
Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Apply the physics

A tone generator plays A at **200 Hz** and B at **400 Hz**. Which has higher pitch?
1. **Find how many cycles each** produces in **2 s**.
2. **Sketch time traces** with equal amplitudes to **show the frequency difference**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Turning up a speaker’s volume does not normally double a song’s pitch. **Explain which wave property mainly** changes.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

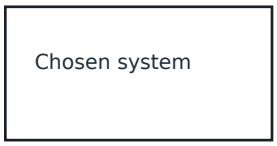
06 Review: connect an earlier idea

A refrigerator uses **24 J** electrical work while moving heat from its colder interior to a warmer room. **Explain why this** is not spontaneous cold-to-hot heat flow.
Label energy entering, leaving and stored.

Word help: spontaneous = able to occur without the specified external work input

Sentence starter: I think … . I can point to …, which shows …, so … .

Draw and label all energy transfers across the boundary.



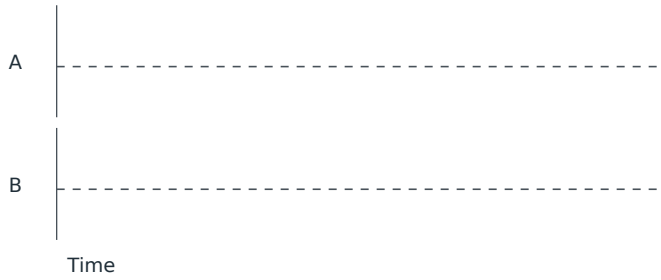
Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Is sound in air mainly transverse or longitudinal? **State how particles move relative** to propagation.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design two sound changes: one changes pitch only and one changes amplitude only.
2. **Describe how their graphs differ**.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3